

A320 takeoff calculator and MCDU uplink takeoff data

The A380 flypad screen for the A32NX, with the A320 FCOM UPLINK TO DATA pages on the MCDU

Branch	feature/a32nx/efb/take-off-calculator
Aircraft	A32NX (and the shared flypad screen of both aircraft)
Commits	b25f790a2 feat(a32nx/efb): A320 takeoff calculator with the MCDU uplink takeoff data
Base	feature/a380/efb/take-off-calculator (document 11)
Status	Merged in develop. Deployed; MCDU pages not yet flown.

Purpose

The same calculator screen for the A320, fed by FBW's A320 performance model, and the takeoff data link of the A320 FCOM between the flypad (the "ground station") and the MCDU.

Specification

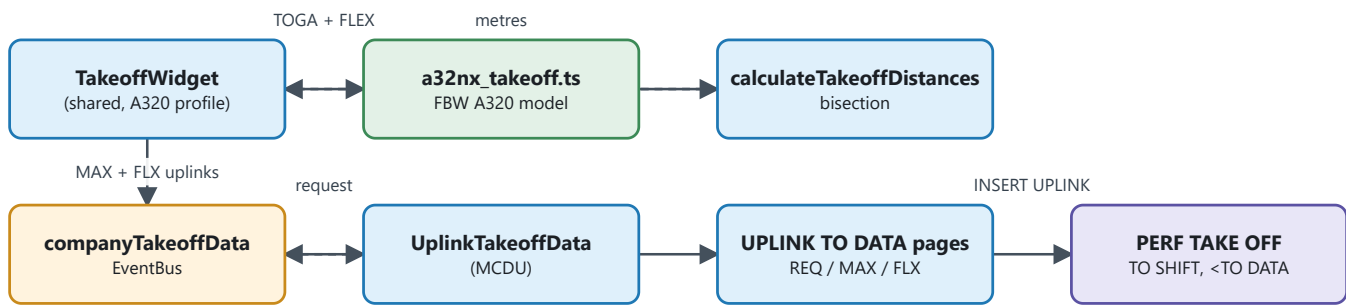
Flypad (shared screen)

- One screen for both aircraft (`TakeoffWidget`, `TakeoffRunway`, `TakeoffFmsLink`); the differences are one profile per aircraft (FMS TOW margins, noise fields and T.O CG only on the A380, text keys).
- Distances from the FBW A320 model: the required length by bisection on the runway length (1000-5000 m), re-running the model; for a FLEX, the length where the model still allows that FLEX.
- The FBW A320 model gives different speeds for TOGA: a separate TOGA calculation feeds the TOGA run and the MAX uplink.
- Performance at forward CG below 27 % MAC (A320 FCOM PER-LOD: forward-CG performance for a CG below 27 %).
- THS shown in the results; V1 / VR / 35 ft estimates share one helper with the A380 (standard atmosphere TAS).
- SEND TO FMS sends two uplinks: MAX (TOGA) and FLEX.

MCDU (A320 FCOM DSC-22_20-50-10-28 P 89-99, DSC-22_45)

- PERF TAKE OFF: `<TO DATA` (6L) opens UPLINK TO DATA REQ, only in the PREFLIGHT and DONE phases on the active plan; the TO SHIFT field works (1 m to the runway length, CLR).
- UPLINK TO DATA REQ (two runways): conditions of the request, sends it; the flypad answers.
- UPLINK MAX TO DATA / UPLINK FLX TO DATA (up to four runways, CONTAM scroll): the received data per runway.
- INSERT UPLINK: only with the runway of the flight plan, and a TOW not more than 1 t below or 3 t above the FMS TOW; inserts V1/VR/V2, FLAPS/THS, FLEX (cleared for MAX), THR RED/ACC, ENG OUT ACC and the shift.
- Messages: TAKEOFF DATA UPLINK, INVALID TAKEOFF UPLINK, REQUEST IS PENDING, NO ANSWER TO REQUEST after 4 min.

 Component added or changed by the feature  MSFS / simulator  Aircraft system (systems host, FMS, FWS)  Display / page
 Data, LVar, storage  Crew, flypad, external program



Architecture

FCOM rules implemented

Rule	Source	Implementation
UPLINK TO DATA REQ for two runways, data received for four	A320 FCOM DSC-22_20-50-10-28 P 89-92	pages + UplinkTakeoffData
Not insertable if the uplink TOW is 3 t greater or 1 t lower than the FMS TOW; runway must match	P 93-96	INSERT UPLINK check
TO SHIFT positive, not greater than the runway length	P 99 [2R]	PERF TAKE OFF field
UPLINK TO DATA prompt in PREFLIGHT and DONE	P 99 [6L]	
TAKEOFF DATA UPLINK / INVALID TAKEOFF UPLINK messages	DSC-22_45	NXSystemMessages
Forward-CG performance below 27 % MAC	A320 FCOM PER-LOD	calculator input

Verification

- tsc, ESLint, Prettier clean; A32NX MCDU and EFB, A380X EFB and MFD build.
- Flypad checked in a headless harness: 70 t on 2500 m gives TOGA 1728 m, FLEX 59 2444 m; labels do not overlap.
- Deployed; the MCDU UPLINK pages have not been seen on screen yet.

PR description (from the PR draft)

From `pr-drafts/12-a320-takeoff-calculator.md`, section "Summary of Changes" onwards.

Summary of Changes

The flypad takeoff screen of PR 11 is shared by both aircraft (`TakeoffWidget`, `TakeoffRunway`, `TakeoffFmsLink`); what differs per aircraft is one profile (FMS TOW margins, noise fields and T.O CG only on the A380, text keys). This PR also completes the takeoff screen and the A380 calculator of PR 11 (see "Both aircraft" and "A380 calculator" below).

A32NX calculator

- Takeoff run from the FBW A320 performance model: the required length by bisection on the runway length (1000-5000 m); for a FLEX, the length where the model still allows that temperature.
- The FBW A320 model gives other speeds for TOGA: a separate TOGA calculation feeds the TOGA run and the MAX uplink.
- Performance at forward CG below 27 % MAC (A320 FCOM PER-LOD). THS in the results.
- SEND TO FMS sends the takeoff data of the thrust selected for the takeoff run only (MAX, or FLEX at its temperature, with that thrust's own speeds).

MCDU (A320 FCOM DSC-22_20-50-10-28 P 89-99, DSC-22_45)

- PERF TAKE OFF: `<TO DATA` (6L) in PREFLIGHT and DONE; the TO SHIFT field (1 m to the runway length, CLR).
- UPLINK TO DATA REQ (two runways), UPLINK MAX TO DATA / UPLINK FLX TO DATA (four runways, CONTAM scroll).
- INSERT UPLINK: runway of the flight plan only, TOW not more than 1 t below / 3 t above the FMS TOW; inserts V1 / VR / V2, FLAPS / THS, FLEX (cleared for MAX), THR RED / ACC, ENG OUT ACC, shift.
- Messages TAKEOFF DATA UPLINK, INVALID TAKEOFF UPLINK, REQUEST IS PENDING, NO ANSWER TO REQUEST (4 min).
- No INSERT UPLINK prompt for an incomplete uplink, e.g. without V1 or VR (A320 FCOM DSC-22_20-70 P 6).

Both aircraft (takeoff screen of PR 11)

- One company takeoff data link for the A380 and A320 FMS (`CompanyTakeoffDataLink`, fbw-common): the request and its 4 min timeout, the data of up to four runways, the common checks and the answer to the flypad import; each aircraft adds its own checks.
- The Data selector (Airbus data only) is removed: with the charts alone the A380 calculator gives no FLEX, V1 or VR and refuses any wind, slope or temperature above TREF. The estimates are always used and stay flagged (amber, asterisk).
- The labels of the missing inputs (heading, TORA, elevation, slope, wind, OAT, QNH, TOW) are amber, so the pilot sees why CALCULATE is not available.
- OFP import: reads the current SimBrief OFP when the flypad has none, handles an OFP without METAR and a runway that is not found, and reports every failure.

A380 calculator (PR 11)

- A TOW of exactly 510 t (the structural MTOW) no longer gives CG OUT OF LIMITS: the envelope limits are inclusive, as on the A320.
- 42 unit tests against the Airbus A380 AC 3-3 chart points and the A380 FCOM (PER-TOF-TOC corrections, PER-TOF-TOR-SRS minimum speeds, PER-TOF-THR-FLX FLEX range, LIM-12 limits).
- The A380X EFB is rebuilt when the calculator changes (EFB `extraDeps`).

Cockpit API Changes

None.

Screenshots (if necessary)

TO ADD: the flypad screen on the A32NX (TOGA and FLEX), the three MCDU pages, PERF TAKE OFF after INSERT UPLINK.

References

- A320 FCOM DSC-22_20-50-10-28 P 89-99 (UPLINK TO DATA pages, PERF TAKE OFF), DSC-22_20-70 P 6 (INSERT UPLINK not displayed for an incomplete uplink), DSC-22_45 (takeoff data function), PER-LOD.
- A380 FCOM PER-TOF-TOC, PER-TOF-TOR-SRS, PER-TOF-THR-FLX, LIM-12; Airbus A380 Aircraft Characteristics (Dec 01/25) 3-3 (tests).

Discord username (if different from GitHub): **TO ADD**

Testing instructions

1. Flypad on the A32NX: calculate, TOGA and FLEX runs; 70 t on 2500 m gives about 1730 m at TOGA.
2. SEND TO FMS; MCDU PERF TAKE OFF `<TO DATA`, then the MAX and FLX pages: INSERT UPLINK fills PERF TAKE OFF.
3. Another runway in the flight plan, or a ZFW 4 t higher: INSERT UPLINK refused.
4. TO SHIFT entry and CLR.
5. SEND TO FMS with TOGA selected, then with a FLEX: only the selected thrust reaches the MCDU (MAX or FLX page).
6. Clear an input (e.g. OAT): its label turns amber and CALCULATE is not available.
7. A380X: TOW 510.0 t with a CG inside the envelope calculates (was CG OUT OF LIMITS); no Data selector on the screen.
8. `npx vitest run fbw-a380x/src/systems/shared/src/performance/a380x_takeoff.spec.ts` (42 tests).